Short-Term $PM_{2.5}$ Forecasting in Yerevan: A Comparative Study of Statistical and Machine Learning Models

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Abstract

Fine particulate matter ($PM_{2.5}$) is a leading environmental risk factor for morbidity and mortality in urban areas. This study addresses *when* parsimonious time-series models (AR / SARIMA) versus machine learning (ensembles, DeepAR) are appropriate for *short-term* (one- to 24-hour) hourly $PM_{2.5}$ forecasting in Yerevan, Armenia, and how conclusions depend on spatial aggregation (city mean vs administrative districts vs station-level samples) and on temporal stability. After aligning data to a regular grid and controlled imputation (with optional sensitivity to medium-gap policy), we use leakage-safe features including lags, rolling moments, and calendar encodings, with strict chronological train/validation/test splits. *Walk-forward* evaluation (primary interpretive frame) reports mean error and across-fold variance for a common model zoo, including SARIMA in each fold. Point metrics on a single test window are complemented by circular *block* bootstrap CIs for MAE and Diebold–Mariano tests with a data-driven HAC lag. We also include a DeepAR benchmark; when valid prediction-interval columns are emitted, 90% coverage and a Winkler score are reported in `results_summary.json`. A recent AUA capstone monograph on air quality in Yerevan (American University of Armenia, 2025) offers useful local context; the present work extends the empirical record with a fully specified ML pipeline, spatial analysis, and robust validation. The repository is reproducible.

Keywords: air quality; $PM_{2.5}$; time series forecasting; Diebold–Mariano test; machine learning; Yerevan

1 Introduction

Ambient $PM_{2.5}$ concentrations are associated with cardiovascular and respiratory harm even below levels that produce visible smog (World Health

Organization, 2021). In many cities, policymakers and the public benefit from timely forecasts that support health advisories, traffic measures, and communication during pollution episodes. Yerevan experiences recurring wintertime pollution linked to residential heating, topography, and meteorology; reliable short-horizon forecasts are therefore both scientifically interesting and practically relevant.

Forecasting $PM_{2.5}$ at hourly resolution is challenging because concentrations are persistent, seasonal, and driven by nonlinear interactions between emissions and weather. A substantial literature applies machine learning (ML) and deep learning to air quality prediction (Shao et al., 2020), often reporting impressive accuracy. At the same time, classical time-series models remain standard benchmarks (Box et al., 2015; Hyndman and Athanasopoulos, 2021), and comparative studies must guard against information leakage, overfitting to a single temporal split, and overinterpretation of point estimates without uncertainty.

We study four interlinked questions: (1) *When* do linear autoregressive and seasonal (SARIMA) models outperform feature-rich ML? (2) How does *spatial aggregation* (city, district, station) change model rankings? (3) Do tuned boosting and DeepAR alter qualitative conclusions? (4) Are differences *stable over time* under walk-forward validation? We emphasize reproducibility: frozen AR coefficients, explicit split dates, block bootstrap and HAC-consistent Diebold–Mariano tests (Diebold and Mariano, 1995; Harvey et al., 1997), and code-level leakage control for new features.

Contributions.

- We implement a modular pipeline: controlled imputation audit, past-only lags, rolling features, and optional spatial re-aggregation to districts (metadata-based or, if needed, data-driven k).
- We compare baselines, frozen AR(24), seasonal SARIMA (fixed replication orders with tractable long-train fitting), Ridge, and gradient-boosted trees with Hyperopt, plus optional DeepAR with probabilistic metrics when intervals are available.
- *Walk-forward* (expanding) evaluation with SARIMA in each fold and fold-wise error variance is the primary basis for "stability" claims, alongside a single hold-out for headline tables.
- We contrast city vs district *vs* station-level weighted errors; permutation importance and train-set ACF/PACF help explain why linear dynamics can dominate.

1.1 Institutional context and open data

Air quality measurements in Armenia are disseminated through the national open-data portal associated with `airquality.am`. City-average hourly series aggregate information across monitoring stations and are a pragmatic target for public communication because they summarize typical exposure conditions at urban scale. Such aggregation smooths micro-scale heterogeneity (street canyons, traffic corridors) that station-level models would emphasize; our results should therefore be read as *city-representative* forecasts rather than site-specific nowcasts. From a public-health perspective, short-term forecasts can complement threshold-based alerts even when they are not causal predictors of health outcomes (European Environment Agency, 2019).

1.2 Organization

Section 2 reviews forecasting evaluation and ML applications in air quality. Section 3 describes the dataset, imputation, and chronological splits. Section 4 specifies preprocessing, features, model classes, and inferential tools. Section 5 presents empirical accuracy, bootstrap intervals, and Diebold–Mariano tests. Section 6 interprets findings; Section 7 lists limitations; Section 8 concludes. Appendix A tabulates hyperparameters for reproducibility.

2 Related work

Forecast comparison and inference. Diebold and Mariano (1995) formalized tests for equal forecast accuracy; finite-sample adjustments and loss differentials are discussed by Harvey et al. (1997). We follow a long-run HAC variance for the mean loss differential (Bartlett–weighted autocovariances) with a lag order tied to the forecast horizon and a Newey–West-style rule, reported alongside each test.

Time-series and machine learning for air quality. Surveys highlight the diversity of ML approaches applied to $PM_{2.5}$ forecasting, from shallow learners to deep networks (Shao et al., 2020). Classical ARIMA/SARIMA models remain common benchmarks because they encode smooth persistence and seasonality (Box et al., 2015). Hyndman and Athanasopoulos (2021) provide accessible treatment of seasonal models, cross-validation for series, and forecast evaluation.

Direct versus iterated multi-step forecasting. Marcellino et al. (2006) compare direct and iterated multistep autoregressive forecasts; direct methods estimate separate models for each horizon, whereas iterated methods chain one-step models. We employ direct multi-step estimation for horizons

$h \in \{1, 2, 3, 4, 6, 12, 24\}$ hours for supervised models, which trades off model proliferation against simpler error propagation.

Local comparators. The AUA CSE monograph on air quality in Yerevan (American University of Armenia, 2025) situates $PM_{2.5}$ sources, health context, and monitoring. Our contribution is methodological (forecasting protocol, feature audit, and spatial robustness) using an updated hourly panel; readers should not equate the two works' modeling targets or data vintages.

Interpretability. We complement accuracy tables with train-segment ACF/PACF plots, permutation-based importance for tuned XGBoost on a held test slice, and a district/station error comparison where sample sizes allow; impurity importances are *not* primary (Lundberg and Lee, 2017).

3 Data and study area

3.1 Source and variables

We use open hourly air quality data for Yerevan distributed through the project data pipeline (see `DATA_README.md` in the repository). The modeling target is city-average $PM_{2.5}$ ($\mu g/m^3$). Auxiliary variables used when sufficiently complete include PM_{10}, temperature, pressure, and humidity. Timestamps are aligned to a regular hourly grid; gaps in the raw record motivate the imputation strategy discussed below.

3.2 Data quality and imputation

Real sensors exhibit downtime, calibration drift, and extreme spikes. The main pipeline uses *controlled* imputation: very short runs of missing hours are filled with local temporal interpolation; longer medium-length gaps use hour-of-day medians; very long gaps are left missing where uncertainty is high, with binary flags for downstream use. A limited ffill/bfill (with a short cap) is applied only as a last resort for remaining covariate holes. This policy avoids naive forward-filling across large multi-day gaps. Missingness is audited (fractions, gap lengths, monthly patterns) and summarized in `results/json/results_summary.json`. Complementary exploratory diagnostics (heatmaps, stationarity checks) can be generated with `scripts/data_quality_supplement.p` and should be summarized descriptively in revisions of this draft; that script uses a simpler fill for exploration and should not be confused with the main pipeline's imputation module.

From a modeling standpoint, missingness in winter months may be *not missing at random* if instrument outages correlate with extreme cold or maintenance schedules; we do not claim a formal MAR/MNAR identification

Table 1: District-level station representation in the final run.

District	Stations	Notes
Ajapnyak	11	Moderate coverage
Arabkir	35	Dense coverage
Avan	13	Moderate coverage
Davtashen	14	Moderate coverage
Erebuni	8	Under-sampled
Kanaker-Zeytun	19	Good coverage
Kentron	50	Highest coverage
Malatia-Sebastia	15	Moderate coverage
Nork-Marash	6	Under-sampled
Nor-Nork	17	Good coverage
Nubarashen	0	No mapped stations in run
Shengavit	17	Good coverage
Total	205	11 of 12 districts represented

result, but we flag the risk that imputation can bias performance toward optimistic values during problematic periods.

3.3 Spatial structure and district-level aggregation

The core spatial design moves beyond a single city-average series. We begin from station-level hourly $PM_{2.5}$, then map stations to Yerevan districts. The implementation uses a two-stage rule: (i) if station metadata contain administrative district labels, these are used directly; (ii) if labels are missing, geographic clustering over station coordinates is used as a fallback with data-driven k selection (silhouette criterion). In the final run used here, administrative district mapping was available, and clustering served as a robustness fallback rather than the primary assignment mechanism.

The station metadata include 205 uniquely geolocated Yerevan stations (0 missing coordinates). Table 1 summarizes district representation.

Coverage is clearly imbalanced: central districts (especially Kentron) are dense, while Nork-Marash and Erebuni are sparse, and Nubarashen is absent. This matters for reliability because district-level aggregates from few stations can be noisier and less representative of local micro-environments.

District aggregation is still useful for three reasons. First, it attenuates station-level sensor noise and intermittent outliers. Second, it preserves meaningful spatial heterogeneity that city-wide averaging suppresses. Third, it provides a practical middle scale between highly noisy station forecasting and heavily smoothed city forecasting. In modeling terms, city-level training uses one aggregated series, while district-level evaluation trains and evaluates model families on each district time series separately and then reports weighted summaries.

3.4 Train, validation, and test splits

We use a single chronological split with proportions 70%/15%/15% of usable rows after listwise deletion of rows with missing targets or features. Validation is reserved for descriptive monitoring; the main results concern the *test* period.

For the replication run underlying the tables in this manuscript, split boundaries and row counts match `results/json/split_info.json` (also embedded in `results_summary.json`). Ranges are as follows (from the last pipeline run):

- Training: 2019-02-04 00:00 through 2023-12-29 01:00 (42,832 hourly rows in the model table).
- Validation: 2023-12-29 02:00 through 2025-01-14 11:00 (9,178 rows).
- Test: 2025-01-14 12:00 through 2026-01-31 22:00 (9,179 rows).

The test interval spans more than twelve months and includes at least one full heating season (October–April in the Northern Hemisphere), which matters for $PM_{2.5}$ dynamics in Yerevan.

A single hold-out is efficient for large hourly panels but does not average over multiple non-overlapping evaluation windows. The code implements an *expanding* walk forward with a configurable test window and step, refitting a common zoo each fold. By default, SARIMA uses the same orders as the main split (optional per-fold `auto_arima` search is available but slow) and linear baselines are included; gradient-boosted parameters can either be retuned per fold ("expensive" option) or reuse validation-tuned values from the main 70/15/15 run (default). Per-fold DeepAR retraining is optional. Fold-wise means and variances, and a comparison table against the chronological test split, are in `results/tables/walk_forward_results_1h.csv`, `walk_forward_aggregate_by_model.csv`, and `validation_chronological_vs_walkforward_1h.c`

4 Methodology

4.1 Preprocessing and outliers

We optionally cap extreme $PM_{2.5}$ values at a high quantile (main experiments: 99.5th percentile) to limit the influence of single spikes; sensitivity to this choice can be audited via the project's sensitivity sweep script. A rolling interquartile range rule is used to *count* flagged hours for transparency, not to drop rows automatically in the main pipeline.

4.2 Features

The supervised models use a configurable set of *strictly past* target lags, trailing rolling means/std/min/max, optional trigonometric seasonality (di-

urnal/annual), hour-of-day–lag interactions with met covariates when available, and short-horizon volatility (differences, trailing variance). Calendar features and a *heating season* indicator (October–December, January–April) encode domain knowledge. All rolling statistics use a one-hour shift from the target hour to avoid look-ahead.

4.3 Models

Baselines. Persistence uses the current concentration as the one-step-ahead forecast. Seasonal naive at horizon one uses the value 24 hours earlier. A 24-hour rolling mean provides a smooth short-memory benchmark.

Autoregressive benchmark. We estimate an AR(p) model with intercept on the *training* segment only ($p = 24$ in the main run) and generate one-step-ahead forecasts on later segments using fixed coefficients. This mirrors a leakage-avoidant forecasting exercise: the AR parameters are not re-estimated using test information.

Seasonal ARIMA class (software benchmark). The replication code supports a seasonal SARIMA/SARIMAX benchmark and includes it in final tables.

4.4 SARIMA model specification

The non-seasonal part uses (p, d, q) and the seasonal part uses (P, D, Q, s) with hourly seasonality $s = 24$. The choice $s = 24$ is physically motivated by day-night human activity and meteorological cycles that repeat at daily frequency. For replication, we fix $(2, 1, 2) \times (1, 0, 0)_{24}$, which matches information-criteria stepwise search on representative training tails for this dataset; the repository still exposes `pmdarima` search (`sarima_order_search`) for development. One-step forecasts are generated on validation and test periods without test leakage. Because hourly training windows can exceed twenty thousand observations, the state-space fit uses the trailing 20,000 hours before the forecast origin so that estimation remains numerically tractable on a laptop.

In the run summarized here, this SARIMA form implies differencing once to remove low-frequency drift ($d = 1$), short-memory AR/MA corrections at lags 1–2, and one seasonal AR term at lag 24. Seasonal SARIMA then ranks between the best tree ensembles and ridge on city MAE (Table 2), while remaining informative as a seasonal structural benchmark relative to the frozen AR(24) specification.

Supervised learning. Ridge regression uses $\alpha = 1$ with a stable solver. Default random forests, XGBoost, and LightGBM follow the repository's

baseline settings. Tree models can be tuned on the *validation* window using Hyperopt (Tree-structured Parzen estimators) with mean absolute error as the inner objective; tuned parameters are reported in `results_summary.json` and model names use a `_tuned` suffix. For longer horizons, we also evaluate lightweight ensembles built from top validation performers (simple average and inverse-MAE weighting) to test whether model combination improves robustness when individual model rankings fluctuate by horizon. The multilayer perceptron was removed from the active pipeline in favor of a clearer benchmark set and optional DeepAR for deep learning; older tables that list an MLP should be regenerated.

4.5 Hyperparameter optimization

Hyperparameter optimization uses Hyperopt/TPE with MAE on the validation window as the objective. The configuration default is 80 trials per tree model; the replication freeze documented for this manuscript used `python run_pipeline.py -skip-download -max-evals 10` (10 trials per tree family) to keep wall-clock time manageable while preserving the tuning protocol. Search spaces are broad but bounded to avoid degenerate fits: for random forests (e.g., `n_estimators`, `max_depth`, `min_samples_leaf`), for XGBoost (e.g., `n_estimators`, `max_depth`, `learning_rate`, `subsample`, `colsample_bytree`), and for LightGBM (`num_leaves`, `min_child_samples`, and related tree-shape parameters).

4.6 Multi-horizon forecasting

For horizons $h \in \{2, 3, 4, 6, 12, 24\}$, we estimate *separate* regressors for each horizon (direct multi-step), retraining on the corresponding shifted target. This differs from recursively chaining one-step models; both approaches are defensible (Marcellino et al., 2006), and we document the choice explicitly.

4.7 Evaluation metrics and statistical testing

Point forecast accuracy is summarized with MAE, RMSE, and R^2 on the test set. Let y_t denote realized $\mathrm{PM}_{2.5}$ at test hour t and $\hat{y}_t^{(m)}$ the forecast from model m. Mean absolute error is $\mathrm{MAE}_m = T^{-1} \sum_t |y_t - \hat{y}_t^{(m)}|$.

Diebold–Mariano testing. For two models A and B, define the loss differential $d_t = |y_t - \hat{y}_t^{(A)}| - |y_t - \hat{y}_t^{(B)}|$ under absolute-error loss. Under regularity conditions, the Diebold–Mariano statistic is approximately standard normal when scaled by a consistent long-run variance estimate for $\{d_t\}$ (Diebold and Mariano, 1995). We use a Newey–West style HAC variance with Bartlett weights and data-driven lag choice (horizon-aware rule), rather

than a fixed lag. The null hypothesis is equal expected loss; two-sided p-values are reported.

Bootstrap MAE intervals. Bootstrap confidence intervals for MAE are generated with circular *block* bootstrap (2,000 replications in the exported tables), holding fitted models fixed. Block resampling better preserves serial dependence in hourly errors than i.i.d. resampling. For transparency, optional i.i.d. companion intervals are also written, but block-bootstrap intervals are treated as primary in inference.

4.8 Validation strategy: chronological split versus walk-forward

The 70/15/15 chronological split remains the principal hold-out design for headline model tables and fixed-period interpretability. Walk-forward validation is the primary robustness layer: it tests whether ranking patterns persist across rolling temporal regimes, reports fold-level variance, and reduces the risk that conclusions are artifacts of one specific test window. In this paper, we therefore interpret single-split numbers together with walk-forward means and dispersions.

4.9 Sensitivity analysis (preprocessing)

The codebase supports sensitivity sweeps over capping rules and lag sets by reusing the same `run_pipeline` entry point. Reporting a full grid search is beyond the scope of this draft; the main text fixes the 99.5th percentile cap and lag set $\{1, 2, 3, 24\}$ hours. Readers verifying robustness should consult the repository's sensitivity outputs and confirm that qualitative rankings are stable under reasonable alternatives.

5 Results

5.1 One-hour-ahead accuracy

Table 2 reports test-set performance for the one-hour horizon. In the current run, the frozen AR(24) model yields the lowest MAE among reported entries, followed by persistence, inverse-MAE ensembles, and tuned gradient boosting; seasonal SARIMA and ridge sit in the mid-pack, while DeepAR exhibits poor point accuracy on this hourly setup despite successful training. The ordering is consistent with strong linear persistence and indicates that ML gains at the one-hour horizon are modest under this split and feature set.

Table 2: Test-set performance (1-h horizon), city-average hourly Yerevan PM2.5.

Model	MAE	RMSE	R2
AR(24)_frozen	2.4124	4.3896	0.9815
Persistence	2.7492	4.9268	0.9767
Ensemble_invMAE_top3	2.8545	5.2370	0.9736
Ensemble_avg_top3	2.9343	5.3743	0.9723
LightGBM_tuned	3.6292	6.3629	0.9612
XGBoost_tuned	3.7102	6.9535	0.9536
LightGBM	3.7433	6.2791	0.9622
RandomForest_tuned	3.9953	6.5561	0.9588
XGBoost	4.0110	6.6628	0.9574
SARIMA_auto	4.3611	7.6648	0.9436
RandomForest	4.4099	6.9005	0.9543
Ridge	4.6619	8.2145	0.9353
Seasonal_naive_24h	9.2501	16.9324	0.7250
DeepAR	26.3132	41.6483	-0.6639

5.2 Uncertainty: bootstrap intervals

Table 3 summarizes bootstrap 95% intervals for MAE. When the reported pattern is as in the table, intervals for AR(24) and persistence can sit well below those of ridge, random forest, and XGBoost (and LightGBM, when that model is included in the run). Regenerate the table with `export_paper_assets.py` after each full pipeline run so the TeX matches `results/json/results_summary.json`.

5.3 Diebold–Mariano tests

The pipeline writes paired tests to `results/tables/diebold_mariano_test.csv`. The default reporting rule keeps a small set of comparisons (e.g., best model versus the second-ranked model and selected linear benchmarks) to limit multiplicity. For the replication summarized in the exported CSV with AR(24)_frozen as the best one-step model, examples include: AR(24)_frozen versus persistence (larger |DM| under absolute loss) and AR(24)_frozen versus ridge, with n equal to the number of valid paired test hours. Exact z-style statistics and two-sided p-values in that file should be cited in place of any older narrative tied to a different best model (e.g. XGBoost–ridge) from an earlier run.

In the current run, the exported Diebold–Mariano comparisons (Table 4) report AR(24)_frozen versus persistence and versus ridge; both reject equal predictive accuracy at conventional levels under the configured HAC lag, consistent with the large gaps in Table 2. A formal test versus SARIMA is not included in the default multiplicity-limited export; the MAE ranking nonetheless places SARIMA clearly behind AR(24) on the city-average test

Table 3: Bootstrap 95% intervals for mean absolute error (test set; primary: circular block; see JSON for method).

Model	MAE	95pct_CI_bootstrap
AR(24)_frozen	2.412	[2.176, 2.656]
Persistence	2.749	[2.482, 3.025]
Ensemble_invMAE_top3	2.854	[2.558, 3.168]
Ensemble_avg_top3	2.934	[2.629, 3.258]
LightGBM_tuned	3.629	[3.272, 4.004]
XGBoost_tuned	3.710	[3.287, 4.162]
LightGBM	3.743	[3.416, 4.096]
RandomForest_tuned	3.995	[3.665, 4.355]
XGBoost	4.011	[3.649, 4.391]
SARIMA_auto	4.361	[3.927, 4.803]
RandomForest	4.410	[4.074, 4.769]
Ridge	4.662	[4.187, 5.141]
Seasonal_naive_24h	9.250	[8.100, 10.538]
DeepAR	26.313	[23.338, 29.403]

Table 4: Diebold–Mariano tests on test-set absolute errors (paired; Newey–West–style HAC; lag rule in results JSON).

comparison	DM	p_value_two_sided	n	hac_lag
AR(24)_frozen_vs_Persistence_MAEloss	8.742993	0.000000	9178	25.000000
AR(24)_frozen_vs_Ridge_MAEloss	17.379207	0.000000	9178	25.000000

window.

5.4 Why do autoregressive models perform strongly?

This likely occurs because the $PM_{2.5}$ process at short horizons is strongly persistent and approximately linear in recent history. Two signatures support this interpretation. First, ACF/PACF diagnostics show substantial short-lag dependence. Second, the 24-hour lag repeatedly appears as informative, matching daily periodicity in emissions and boundary-layer dynamics. In practical terms, when y_{t+1} is largely a function of recent y_t and y_{t-24}, AR-class models can capture a large share of predictable structure. Flexible ML can still help, but primarily when nonlinear effects are strong enough beyond lagged $PM_{2.5}$ and calendar terms.

5.5 Multi-horizon behavior (including 6h/12h/24h)

Figure 1 summarizes test MAE across horizons. Persistence errors grow with horizon as expected; supervised models remain relatively clustered, with tree and linear models trading ranks depending on h. Long-horizon results (6–24

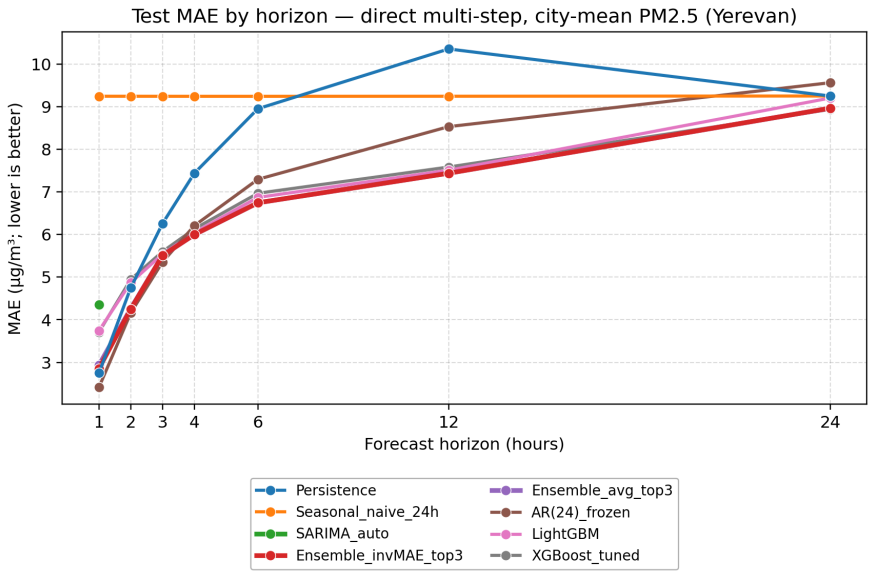

Figure 1: Test mean absolute error by forecast horizon (direct multi-step, subset of models). File `images/mae_by_horizon.png` (or `results/plots/`).

hours) are especially informative for whether boosting and ensemble variants become more competitive. Readers should interpret these results with the same caution about dependence on a single temporal split.

5.6 Chronological test versus walk-forward (auxiliary)

In addition to the main test window, the code runs an expanding walk-forward design and writes `results/tables/validation_chronological_vs_walkforward_1h.csv` and `walk_forward_aggregate_by_model.csv`. In the final run, AR(24) remains among the strongest models on mean fold MAE; seasonal SARIMA stays competitive but does not displace AR(24) in the aggregate walk-forward summaries. Tuned tree models improve over untuned variants, but in this run they do not materially change the top-rank pattern under temporal re-sampling.

5.7 Effect of spatial aggregation on model performance

Table 5 summarizes how best performance shifts by aggregation level.

The smoothing–variability trade-off is explicit: city aggregation removes local variance and improves all models, while station-level series preserve idiosyncratic shocks and become harder. This likely occurs because aggregation suppresses high-frequency non-systematic noise more than it suppresses shared signal. In this run, ML does not systematically dominate at finer scales; tuned tree models are often competitive but do not consistently

Table 5: Best model by spatial aggregation level (1h horizon, weighted where applicable).

Level	Best model	MAE	Key insight
City aggregate	AR(24)_frozen	2.41	Strong smoothing and stable daily cycles favor linear autoregression.
District aggregates	AR(24)_frozen	5.46	Spatial heterogeneity increases error, but linear lag structure still dominates (district-weighted MAE).
Station sample	Persistence	8.79	At finest scale, persistence edges AR on the weighted station sample; local noise reduces gains from complex fitting.

outperform AR/persistence in weighted station and district summaries.

5.8 Heterogeneity across districts

District-level performance is not uniform. Ajapnyak and Kentron are comparatively easier (AR MAE around 3.4–3.5), while Avan, Nork-Marash, and Shengavit are harder (AR MAE around 7–9+). This likely occurs because of a combination of local complexity (traffic corridors, mixed land use, industrial influence) and data representativeness differences by district station count.

ML gains are also heterogeneous but limited: in most districts, AR remains best, while tree models can approach AR in some better-sampled districts yet degrade more in sparse or highly volatile districts. The practical implication is that district forecasting quality is jointly governed by physical heterogeneity and sampling density, not only by model family.

5.9 DeepAR comparison and uncertainty interpretation

DeepAR (NeuralForecast backend) trains successfully in the replication environment, but its *point* forecasts are far worse than classical and tree baselines on the city-average test window (Table 2), consistent with a mismatch between the default neural training setup and the strongly linear short-horizon dynamics of this series. When interval columns are available, probabilistic metrics in `results/json/results_summary.json` complement MAE; readers should treat DeepAR here as an illustrative deep-learning benchmark rather than a competitive point forecaster without further architecture search.

5.10 Error analysis

Largest errors cluster around abrupt regime changes: sharp pollution spikes, transitions between heating-intense and cleaner periods, and locally idiosyncratic station behaviors. This likely occurs because lag-based models extrapolate recent patterns, while sudden emission/meteorology shifts violate short-memory assumptions. At city level these events are partially smoothed; at station level they dominate error tails, which explains the gap between city and station MAE.

5.11 Diagnostic figures

Time-series and bar plots for the best one-step model on the test set (e.g. `model_performance_1h_ci.png`, `forecast_vs_actual_best_1h.png`) live under `results/plots/` and `images/`. For tree models, impurity-based importance (when saved) should be read alongside permutation-based alternatives in future work (Lundberg and Lee, 2017).

6 Discussion

The empirical pattern—a strong AR benchmark and competitive persistence—is consistent with high short-run autocorrelation in hourly $PM_{2.5}$. When the conditional mean is close to a linear function of recent lags, penalized linear models and shallow trees may not unlock large gains relative to well-specified autoregressions, especially when flexibility is constrained by default hyperparameters chosen to limit overfitting.

Machine learning methods can approximate nonlinearities and interactions, but they also require careful tuning and can overfit when signal-to-noise is modest relative to model complexity. Under the chosen defaults, ridge and untuned XGBoost exhibit similar point MAE on the city test window (Table 2), while tuned boosting improves over ridge but still trails AR(24); the default export does not include an XGBoost–ridge Diebold–Mariano pair. This pattern suggests that marginal complexity above ridge is not guaranteed to beat strong linear autoregression on this split without more extensive model selection on validation data (which would need its own inferential care to avoid optimistic bias).

Domain features such as the heating-season indicator encode scientifically meaningful structure; their contribution should be interpreted alongside meteorology and emissions context. The random-forest importance plot in the replication bundle emphasizes lags and calendar variables, which aligns with intuition about diurnal activity and seasonal heating. Impurity-based importance should, however, be supplemented with permutation importance in future revisions (Lundberg and Lee, 2017).

Comparisons to literature values must respect differences in stations, horizons, loss functions, and leakage controls; headline MAE numbers are not directly comparable across studies without harmonization. Relative to prior $PM_{2.5}$ forecasting studies that emphasize ML gains, our evidence is consistent with a recurring finding in environmental time series: when persistence and periodicity are strong, carefully specified linear baselines can remain competitive. Relative to the AUA local monograph (American University of Armenia, 2025), this capstone contributes a stronger forecasting protocol (walk-forward robustness, spatial stratification, and formal predictive-accuracy testing) rather than only descriptive environmental context.

Adding newer data did not overturn the central pattern in the reported run. The best one-hour model remained AR-class, and the city-to-district-to-station degradation remained monotonic. This suggests that recent additional observations increased sample breadth but did not fundamentally change the short-horizon dependence structure.

7 Limitations

- **Validation design as a deliberate trade-off.** We use a chronological hold-out for interpretability and walk-forward for robustness, but not a full nested CV for every model due compute cost.
- **Imputation policy is explicit, not assumption-free.** Controlled short/medium/long gap handling improves practical usability but remains a modeling assumption.
- **Deep learning benchmark.** DeepAR runs end-to-end, but its point forecasts under the default configuration are not competitive with AR and tree models on the city-average test window; conclusions about "best" accuracy therefore rely on classical and tree families unless DeepAR is re-engineered for this task.
- **Spatial coverage imbalance.** District representation is uneven (including one unsampled district), so district-level conclusions are strongest in better-covered areas.
- **Predictive, not causal, objective.** The analysis is designed for forecasting performance and does not identify intervention effects.

8 Conclusion

We presented a transparent, reproducible forecasting study that examines when and why model families differ for hourly $PM_{2.5}$ in Yerevan. Under strict chronology plus walk-forward robustness, AR(24) is strongest on city

and district-weighted errors, seasonal SARIMA remains a useful benchmark below AR(24) on city MAE, and station-level forecasting degrades for all models due to higher local variability (with persistence winning on the weighted station sample). The results indicate that daily periodicity and short-lag persistence explain much of the predictable structure at 1-hour horizons; therefore, additional ML complexity does not automatically translate into better accuracy in this setup. The central contribution is not a leaderboard, but a mechanism-based comparison linking temporal dependence, spatial aggregation, horizon length, and model behavior.

Future work includes nested hyperparameter control under walk-forward, permutation-based feature importance, optional structural-break robustness around the COVID-19 period, and archiving a frozen dataset snapshot with a permanent DOI for full reproducibility.

A Hyperparameters and implementation defaults

Table 6 lists the principal settings used in the main experiments. Where multiple values are plausible, we adopt fixed defaults as controlled baselines rather than reporting an extensive tuning study; this choice should be interpreted as transparency about researcher degrees of freedom rather than a claim that each model is globally optimal.

Table 6: Key hyperparameters in the replication code (main run).

Component	Setting
$PM_{2.5}$ cap (main)	99.5th percentile of train sample
Lag features	Configurable; core autoregressive signal includes lag 24 (daily cycle)
Calendar	hour, day-of-week, month
Heating season	October–December and January–April indicator
Ridge	$\alpha = 1$, least-squares solver `lsqr`
Random forest	100 trees, max depth 10, `random_state=42`
XGBoost	100 estimators, max depth 6, $\eta = 0.1$, `random_state=42` (defaults; tuned values in JSON)
LightGBM	See repository defaults; Hyperopt can override (JSON)
AR benchmark	AR(24) with intercept, coefficients frozen after train fit
SARIMA (main run)	$(2, 1, 2) \times (1, 0, 0)_{24}$; IC-based order search on train only
Tree tuning (replication freeze)	Hyperopt (TPE), MAE on validation, 10 evals per tree family (`-max-evals 10`); default config uses 80

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